# Network Mini-project

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We study the structural properties of the all-or-none network growth model, a variation of preferential attachment and copying mechanisms in which a new node connects to a uniformly chosen target and, with probability $p$, to *all* of its neighbours, or with probability $1-p$ to *none*. We derive analytical expressions for the expected number of edges as a function of $p$ and the number of nodes $N$, characterise the degree distribution in the sparse regime, and compute the clustering coefficient, assortativity coefficient, and average shortest-path distance. Our analytical predictions are supported by extensive numerical simulations.

## I. INTRODUCTION

Network growth models are central to our understanding of real-world complex systems, ranging from the World Wide Web to biological and social networks [1, 2]. The Barabási–Albert (BA) model [1] introduced preferential attachment as a mechanism for generating power-law degree distributions, while copying models [3] propose an alternative microscopic rule in which new nodes replicate the connections of existing ones.

In this work we analyze the *all-or-none* model: at each growth step a new node is added and connected to a target chosen uniformly at random; it then connects to *all* neighbours of the target with probability $p$, or to *none* of them with probability $1-p$. This binary choice creates a sharp interpolation between two qualitatively distinct regimes and leads to non-trivial structural properties that we characterise below.

The remainder of this paper is organised as follows. Section II defines the model precisely. Sections III–VII derive analytical results and present numerical evidence for the edge count, degree distribution, clustering, assortativity, and average shortest-path distance, respectively. Section VIII concludes.

## II. MODEL DEFINITION

The network begins with a single node ($N = 1$). At each growth step ($N \to N+1$):

1. A new node $v$ is introduced.
2. A target node $u$ is chosen uniformly at random from the existing $N$ nodes.
3. An edge $(v, u)$ is added.
4. With probability $p$, edges $(v, w)$ are added for every neighbour $w$ of $u$; with probability $1-p$, no further edges are added.

## III. NUMBER OF EDGES

Let $M_N$ denote the number of edges in the all-or-none network of $N$ nodes. At step $N \to N+1$, a target $u$ is chosen uniformly from the $N$ existing nodes. The expected number of new edges is $1 + p\,k_u$, where $k_u$ is the degree of $u$. Note that by the handshake lemma, the average value of $k_u$ over all choices of $u$ is $2M_N/N$. It follows that,

$$\begin{aligned}\mathbb{E}[M_{N+1}] &= \mathbb{E}[M_N] + \left(\frac{2p}{N}\mathbb{E}[M_N] + 1\right) \\ &= \left(1 + \frac{2p}{N}\right)\mathbb{E}[M_N] + 1. \qquad (1)\end{aligned}$$

This is a first-order linear recurrence. A particular solution of the form $\alpha N$ yields $\alpha = 1/(1-2p)$ for $p \neq 1/2$. The homogeneous solution is $A_N = \Gamma(N+2p)/[\Gamma(1+2p)\,\Gamma(N)]$, satisfying $A_{N+1}/A_N = (N+2p)/N$. Imposing $\mathbb{E}[M_1] = 0$ now yields

$$\boxed{\mathbb{E}[M] = \frac{1}{1-2p}\left(N - \frac{\Gamma(N+2p)}{\Gamma(1+2p)\,\Gamma(N)}\right)}, \quad p \neq \tfrac{1}{2}. \quad (2)$$

Since $\Gamma(N+2p)/\Gamma(N) \sim N^{2p}$ for large $N$, two regimes emerge:

- *Sparse* ($p < 1/2$): $\mathbb{E}[M] \sim N/(1-2p)$ (linear).
- *Dense* ($p > 1/2$): $\mathbb{E}[M] \sim N^{2p}/[(2p-1)\,\Gamma(1+2p)]$ (superlinear).

### A. Numerical results

FIG. 1. Expected number of edges $\mathbb{E}[M]$ as a function of $p$ for several values of $N$, obtained by averaging over $10^4$ independent realisations. Solid lines show the analytical prediction (2).

## IV. DEGREE DISTRIBUTION

Denote by $f(k, i, N)$ the probability that node $i$ has degree $k$ when the network has $N$ nodes. At step $N \to N+1$, degree $k$ increases by one with probability $(1+pk)/N$, because either the node is chosen as target (probability $1/N$) or a neighbour of it is chosen and the

"all" event occurs (probability $pk/N$) The master equation for $f$ is then

$$f(k,i,N+1) = \frac{1+p(k-1)}{N}\, f(k-1,i,N) + \left(1-\frac{1+pk}{N}\right) f(k,i,N). \quad (3)$$

When $N$ is large, we seek the asymptotic degree distribution $P_k = \lim_{N\to\infty} n_k(N)/N$, where $n_k(N) = \sum_{i=1}^{N} f(k,i,N)$ is the expected number of nodes with degree $k$ in the $N$-node network. Note that the new node enters with degree 1 with probability $1-p$ and with degree $k$ with probability $p\, n_{k-1}(N)/N$ for $k \geq 2$. Hence, summing (3) over the $N$ existing nodes and adding the new node yields

$$n_k(N+1) = \frac{1+p(k-1)}{N}\, n_{k-1}(N) + \left(1-\frac{1+pk}{N}\right) n_k(N) + (1-p)\,\delta_{k,1} + \frac{p\, n_{k-1}(N)}{N}. \quad (4)$$

Substituting $n_k(N) = NP_k$, we have

$$(2+pk)\, P_k = (1+pk)\, P_{k-1} + (1-p)\,\delta_{k,1}. \quad (5)$$

This yields $P_1 = (1-p)/(2+p)$ and, for $k \geq 2$,

$$P_k = \frac{1+pk}{2+pk}\, P_{k-1}, \quad (6)$$

which have the closed form

$$\boxed{P_k = \frac{1-p}{2+p} \prod_{j=2}^{k} \frac{1+pj}{2+pj}}, \quad k \geq 1. \quad (7)$$

In terms of the Gamma function,

$$P_k = \frac{(1-p)\,\Gamma(2+2/p)}{(2+p)\,\Gamma(2+1/p)} \cdot \frac{\Gamma(k+1+1/p)}{\Gamma(k+1+2/p)}. \quad (8)$$

For large $k$, $\Gamma(k+1+1/p)/\Gamma(k+1+2/p) \sim k^{-1/p}$, so

$$P_k \sim c_p\, k^{-1/p}, \quad k \to \infty, \quad (9)$$

where $c_p = (1-p)\,\Gamma(2+2/p)/[(2+p)\,\Gamma(2+1/p)]$. The degree distribution thus follows a power law with exponent $1/p$, which matches the degree distribution of the BA model at $p = 1/3$.

## V. TRIANGLES AND CLUSTERING

### A. Triangles

Let $T_N$ denote the number of triangles in the all-or-none network of $N$ nodes. Triangles are created if and only if the "all" event occurs and $N \geq 2$. If the target $u$ has degree $k_u$ and there are $e_u$ edges among its neighbours, the new node forms $k_u$ triangles of type $(v_{\text{new}}, u, w_i)$ and $e_u$ triangles of type $(v_{\text{new}}, w_i, w_j)$. Note that $\sum_u k_u = 2M_N$ and $\sum_u e_u = 3T_N$. Averaging over the target choice $u$ thus yields

$$\mathbb{E}[T_{N+1}] = \left(1+\frac{3p}{N}\right)\mathbb{E}[T_N] + \frac{2p\,\mathbb{E}[M_N]}{N}. \quad (10)$$

This linear recurrence has the same structure as (1). Substituting the exact edge formula and solving with $\mathbb{E}[T_1] = 0$ gives, for $p \neq 1/2, 1/3$,

$$\mathbb{E}[T] = \frac{2pN}{(1-2p)(1-3p)} + \frac{2A_N}{1-2p} - \frac{2B_N}{1-3p}, \quad (11)$$

where $A_N = \Gamma(N+2p)/[\Gamma(1+2p)\Gamma(N)]$ and $B_N = \Gamma(N+3p)/[\Gamma(1+3p)\Gamma(N)]$.

### B. Clustering coefficient

The global clustering coefficient is

$$C = \frac{\#\text{triangles}}{\#\text{connected triples}/3}.$$

Note that the number of connected triples is $\frac{1}{2}\sum_i k_i(k_i-1)$. Tracking $S_N = \sum_i k_i^2$ via a recurrence analogous to (10),

$$S_{N+1} = \left(1+\frac{3p}{N}\right) S_N + 2 + \frac{(4+6p)M_N}{N}, \quad (12)$$

and using $W = (S-2M)/2$, one finds that the dominant growth rates of $3T$ and $W$ share the same exponent for all $p$, and their ratio converges to a $p$-dependent constant. Specifically, for $p < 1/3$ both $T$ and $W$ grow linearly in $N$, while for $p > 1/3$ both grow as $N^{3p}$. In either case the prefactors yield

$$\boxed{C = \frac{3p}{1+2p}}, \quad (13)$$

valid for all $p \in (0,1)$ in the large-$N$ limit. This is *independent of* $N$, a hallmark of small-world networks. The formula interpolates from $C = 0$ (tree at $p = 0$) to $C = 1$ (complete graph at $p = 1$), with $C = 3/7 \approx 0.43$ at $p = 1/3$.

FIG. 2. Global clustering coefficient $C$ as a function of $p$ for $N = 10^3$. Error bars represent one standard deviation over $10^3$ realisations. The dashed line is the analytical prediction (13).

## VI. ASSORTATIVITY

The degree assortativity coefficient [4] can be written as

$$r = \frac{4MR - S^2}{2MK_3 - S^2}, \quad (14)$$

where $M = M_N$, $S = \sum_i k_i^2$, $K_3 = \sum_i k_i^3$, and $R = \sum_{(i,j)\in E} k_i k_j$. We already have recurrences for $M$ and $S$; a closed-form for $r$ requires tracking $K_3$ and $R$ via the same moment-closure method.

### A. Recurrences for higher moments

When node $v_{\text{new}}$ is added, the target $u$ gains degree 1 (always) and each neighbour $w_i$ of $u$ gains degree 1 (with probability $p$, all or none). Accounting for degree changes of existing nodes and the new node's contribution, one obtains

$$K_{3,N+1} = \left(1 + \frac{4p}{N}\right) K_{3,N} + 2 + \frac{(3+6p)S_N + (6+8p)M_N}{N}, \tag{15}$$

$$R_{N+1} = \left(1 + \frac{4p}{N}\right) R_N + 1 + \frac{(1+3p)S_N + (2+6p)M_N + 3pT_N}{N}. \tag{16}$$

The derivation of (16) uses the identity $\sum_u \sum_{w\sim u} \sigma_w = 2R$, where $\sigma_w = \sum_{z\sim w} k_z$, since each edge $\{w, z\}$ contributes $k_w k_z$ to both sides via the factor $k_w$.

Both recurrences share the homogeneous growth factor $(1 + 4p/N)$, with solution $\Phi_N = \Gamma(N + 4p)/[\Gamma(1 + 4p)\,\Gamma(N)] \sim N^{4p}/\Gamma(1 + 4p)$.

### B. Closed-form in the sparse regime

For $p < 1/4$, all quantities $M, S, T, K_3, R$ grow linearly: $X_N \sim \alpha_X N$. Substituting this ansatz into (15)–(16) and using the known coefficients $\alpha_M$, $\alpha_S$, $\alpha_T$ gives

$$\alpha_{K_3} = \frac{26 + 22p}{(1-2p)(1-3p)(1-4p)}, \quad \alpha_R = \frac{3(3+5p)}{(1-2p)(1-3p)(1-4p)}. \tag{17}$$

Inserting into (14), the factors $(1-2p)$, $(1-3p)$, and $(1-4p)$ cancel between numerator and denominator, yielding

$$\boxed{r = \frac{p(9-2p)}{(1+2p)(2-p)}}, \quad p < \tfrac{1}{3}. \tag{18}$$

This is derived from the exact algebraic identity $4\alpha_M\alpha_R - \alpha_S^2 = 8p(9-2p)(1-p)/[\cdots]$ and $2\alpha_M\alpha_{K_3} - \alpha_S^2 = 8(1-p)(1+2p)(2-p)/[\cdots]$, with common denominators cancelling.

The formula gives $r(0) = 0$ (uncorrelated random recursive tree) and increases monotonically to $r \to 1$ as $p \to 1/3^-$. The strong assortativity near $p = 1/3$ reflects the divergence of $\langle k^2\rangle$: a few high-degree hubs become mutually connected through repeated "all" events, dominating the degree–degree correlation. For $p \geq 1/3$, the linear-growth regime breaks down and the homogeneous terms ($\sim N^{4p}$) in $K_{3,N}$ and $R_N$ dominate; numerical simulations indicate that $r$ decreases for $p > 1/3$.

FIG. 3. Assortativity coefficient $r$ as a function of $p$ for $N = 10^3$. The dashed line is the analytical prediction (18).

## VII. AVERAGE SHORTEST-PATH DISTANCE

Unlike the previous quantities, the average shortest-path distance $\ell$ can be computed via an *exact* recurrence on the total sum of pairwise distances $D_N = \sum_{i<j} d(i, j)$, without any mean-field approximation.

### A. Exact distance recurrence

The key observation is that adding $v_{\text{new}}$ never shortens any existing pairwise distance (any path through $v_{\text{new}}$ can be rerouted through the target $u$ or its neighbours). When the "all" event occurs, $v_{\text{new}}$ acquires the *same closed neighbourhood* as $u$, so for every existing node $j \neq u$, the first step on a shortest $u$–$j$ path goes through some neighbour $w$ of $u$; since $v_{\text{new}} \sim w$, we obtain $d(v_{\text{new}}, j) = 1 + d(w, j) = d(u, j)$ exactly. Together with $d(v_{\text{new}}, u) = 1$:

$$\begin{aligned} \text{“all”}:&\ \textstyle\sum_j d(v_{\text{new}}, j) = 1 + \sigma_u, \\ \text{“none”}:&\ \textstyle\sum_j d(v_{\text{new}}, j) = N + \sigma_u, \end{aligned} \tag{19}$$

where $\sigma_u = \sum_{j\neq u} d(u, j)$ and the sums run over the $N$ existing nodes. Averaging over the uniform choice of $u$ and the binary event,

$$\mathbb{E}[D_{N+1}] = \left(1 + \frac{2}{N}\right) \mathbb{E}[D_N] + (1-p)N + p, \tag{20}$$

using $\frac{1}{N}\sum_u \sigma_u = 2D_N/N$. This recurrence is *exact in expectation*, with $\mathbb{E}[D_1] = 0$.

### B. Asymptotic solution

The homogeneous solution is $\mathcal{H}_N = N(N + 1)/2$. By variation of parameters, $\mathbb{E}[D_N] = \mathcal{H}_N\, c_N$ with partial fractions giving

$$c_N = 2(2p-1)(H_N - 1) + 2(2-3p)\left(H_{N+1} - \tfrac{3}{2}\right), \tag{21}$$

where $H_N = \sum_{j=1}^N 1/j$ is the harmonic number. For large $N$, $c_N \sim 2(1-p)\ln N$, hence $\mathbb{E}[D_N] \sim (1-p)N(N+1)\ln N$. Dividing by $\binom{N}{2}$ pairs,

$$\boxed{\ell(N) \sim 2(1-p)\ln N}, \tag{22}$$

valid for all $p \in [0, 1)$. At $p = 0$ this recovers the known Wiener-index scaling $\ell \sim 2\ln N$ of the random recursive tree; at $p = 1$ the network is $K_N$ with $\ell = 1$.

The distance scales *logarithmically* for every $p < 1$, with no phase transition. The high clustering ($C = O(1)$,

Section V) invalidates the locally tree-like assumption underlying the branching-process heuristic $\ell \sim \ln N/\ln\kappa$, which would incorrectly predict bounded distances for $p > 1/3$. The exact recurrence (20) bypasses this difficulty entirely.

FIG. 4. Average shortest-path distance $\ell$ as a function of $N$ on a log-linear scale for several values of $p$. Dashed lines show the analytical prediction $2(1-p)\ln N$.

## VIII. CONCLUSION

We have studied the all-or-none network growth model and derived exact or asymptotic analytical expressions for five structural properties, all controlled by the single parameter $p$:

(i) The expected number of edges transitions from linear ($\sim N/(1-2p)$ for $p < 1/2$) through $N \ln N$ at $p = 1/2$ to superlinear ($\sim N^{2p}$) for $p > 1/2$. (ii) The degree distribution follows a power law $P_k \sim k^{-1/p}$ with a continuously tunable exponent, matching the independent copying model at the mean-field level. (iii) The global clustering coefficient converges to $C = 3p/(1+2p)$, a remarkably simple $N$-independent expression valid for all $p$. (iv) In the sparse regime ($p < 1/3$), the assortativity coefficient is $r = p(9-2p)/[(1+2p)(2-p)]$, derived by moment closure, increasing from $r = 0$ at $p = 0$ to $r \to 1$ as $p \to 1/3^-$. (v) An exact topological recurrence for the sum of pairwise distances shows that the average shortest-path distance scales as $\ell \sim 2(1-p)\ln N$ for all $p < 1$, with no phase transition.

The network is a *small world* for every $p \in (0,1)$: logarithmic distances coexist with $O(1)$ clustering. The parameter $p$ smoothly interpolates between a random recursive tree ($p = 0$) and the complete graph ($p = 1$), passing through regimes that reproduce key features of real-world networks including scale-free degree distributions, tuneable assortativity, and the small-world property. Future directions include extending the assortativity formula beyond $p = 1/3$, analysing the model on directed or weighted networks, and studying its spectral properties.

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